From a paper algorithm to Python

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for From a paper algorithm to Python before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

> Four packs contain six counters each. What steps find and show the total?

My starting explanation

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported From a paper algorithm to Python trace and record the first step you can explain without help.

Use W2: Predict then observe for From a paper **algorithm** to Python. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed From a paper algorithm to Python explanation against the supplied facts before discussing it.

Use the worked case for From a paper **algorithm** to Python. Proposed explanation: “An equals sign displays the answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Trace variables and a change

Predict the printed value. Change only packs to 5 and predict again. Explain which statements apply a **written rule**.

```
packs = 3
per_pack = 7
total = packs * per_pack
print(total)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Connect the paper steps to Python

> Connect the paper steps to Python. Apply the supplied From a paper algorithm to Python facts and keep your reasoning.

> Write the complete four-line script for 4 packs of 6. Beside each line explain what it does. Which line displays information?

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Predict then observe

> Predict then observe. Apply the supplied From a paper algorithm to Python facts and keep your reasoning.

Make records for 4 packs of 6, then 5 packs of 6. Predict before running. Record actual **output**, any error and what changed. If working on paper, label both results as traces.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Explain an order error

> Explain an order error. Apply the supplied From a paper algorithm to Python facts and keep your reasoning.

Move print(total) to the first line of a fresh script. Predict what happens and why. Put the lines back into an order that meets the requirement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Transfer the procedure independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Transfer the procedure independently. Apply the supplied From a paper algorithm to Python facts and keep your reasoning.

A fictional shelf has 3 trays of 7 tiles each. Write a four-line Python script using trays, per_tray and tiles. Predict its **output**. Then change only trays to 0 and predict again. Explain why adding the two quantities is wrong.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Keep an honest conclusion

> Keep an honest conclusion. Apply the supplied From a paper algorithm to Python facts and keep your reasoning.

Explain the difference between a successful run and a correct calculation. State one limit of this **program** and what you would keep so another person can check your work.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect From a paper algorithm to Python to Python readiness portfolio, identify a limit and choose the next useful test.

Add a labelled from a paper **algorithm** to python **evidence** sheet to Python readiness portfolio. Show how you can translate a short paper calculation into a python script.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: From a paper algorithm to Python

> Convert a seed-count algorithm. The paper plan counts seeds in identical trays, then adds a labelled spare tray. Explain the labelled step before checking the changed condition.

Convert a seed-count **algorithm**. The paper plan counts seeds in identical trays, then adds a labelled spare tray.

Python 3: trays=4; per_tray=6; spare=2; total=trays*per_tray+spare; print(total). All quantities count seeds.

AI ACADEMY / PYTHON-BRIDGE/1

From a paper algorithm to Python

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name the inputs

Four trays hold six seeds each; two extra seeds are separate.



2. Translate the plan

Multiply tray count by seeds per tray, then add the spare count.



3. Trace the assignment

4*6+2 is 26; total stores 26 and print displays it.

CHANGE ONE THING

Change spare to 5 while keeping both tray inputs fixed.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Match each paper quantity to its Python name. Why does the spare count get added once, and what prints when spare is five?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation.

Use actual Python 3 results to check the **prediction** after reasoning first.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Python readiness portfolio record.

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	

Evidence to keep	My record
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Convert a seed-count **algorithm**. The paper plan counts seeds in identical trays, then adds a labelled spare tray.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? 4*6+2 is 26; total stores 26 and print displays it.

> 1. Trace the **assignment**

2. Name the **input**s

3. Translate the plan

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Trace the **assignment** 4*6+2 is 26; total stores 26 and print displays it. Python follows the arithmetic instructions.

This translation automates a declared counting rule; it does not train a model from examples.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change spare to 5 while keeping both tray **input**s fixed. Which conclusion is supported by the supplied facts?

1. The printed total becomes 29.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The printed total becomes 29. 4*6 remains 24; adding five gives 29.

Repair a missing building block

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

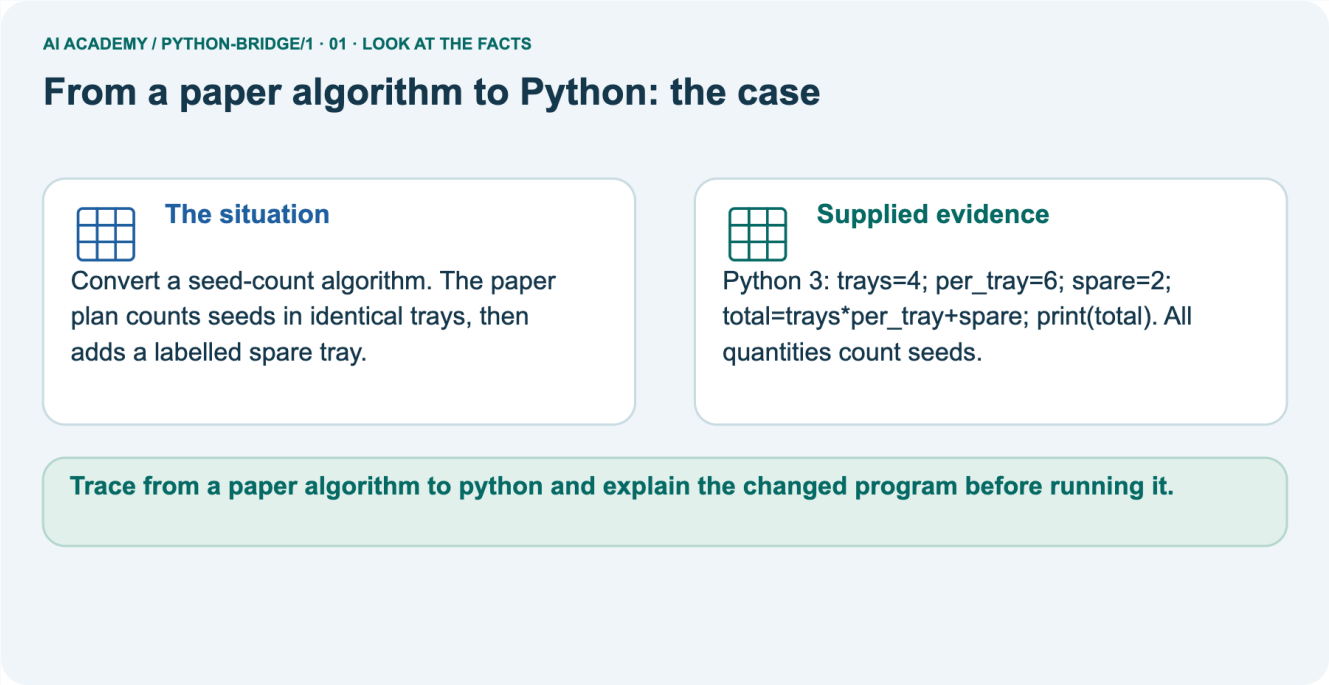
> **What I repaired, the evidence I kept and the next test**
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> | |
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> | |
> | |
> | |

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each From a paper algorithm to Python picture from its labels, then record a relevant limit.

01. From a paper algorithm to Python: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

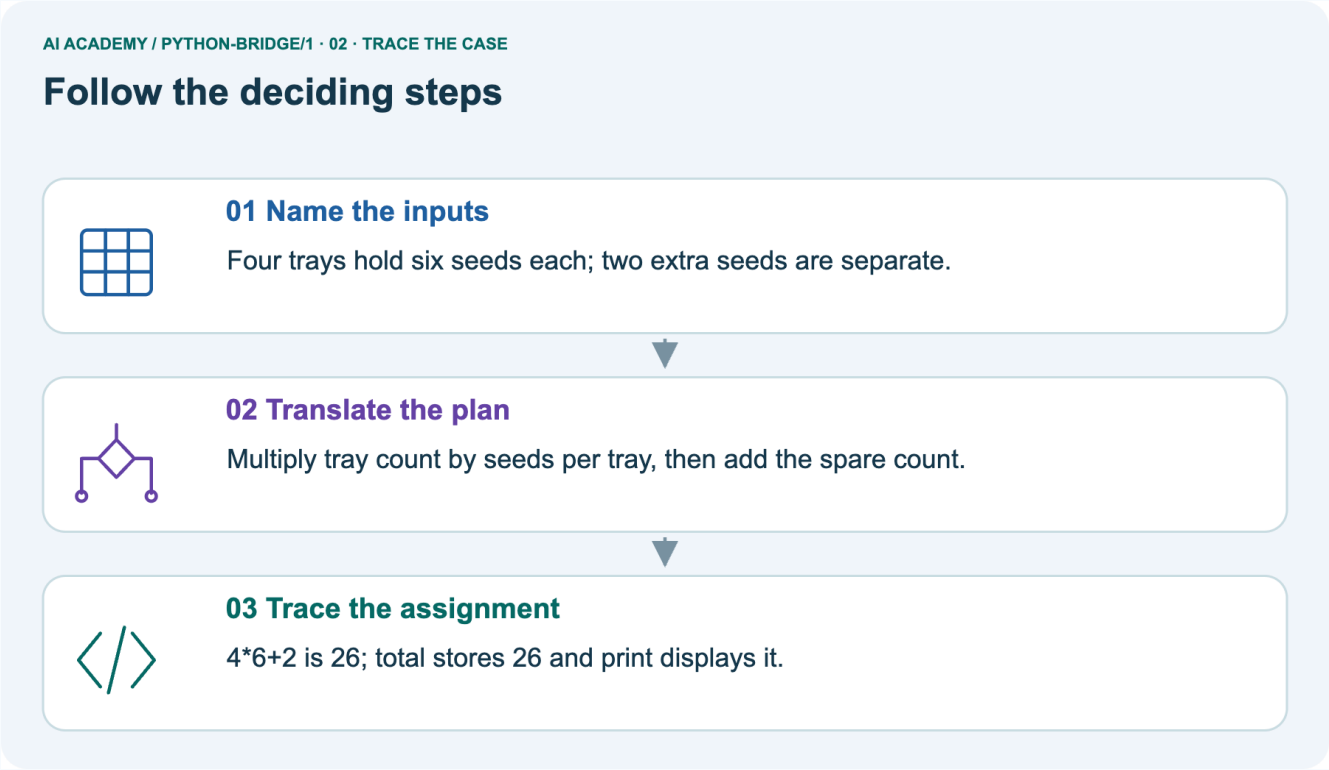
Convert a seed-count **algorithm**. The paper plan counts seeds in identical trays, then adds a labelled spare tray. Supplied facts: Python 3: trays=4; per_tray=6; spare=2; total=trays*per_tray+spare; print(total).

All quantities count seeds.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Name the **input**s: Four trays hold six seeds each; two extra seeds are separate. → Translate the plan: Multiply tray count by seeds per tray, then add the spare count. → Trace the **assignment**: 4*6+2 is 26; total stores 26 and print displays it.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A calculation from paper to output

AI ACADEMY / PYTHON-BRIDGE/1 · 03 · INSPECT THE LESSON MODEL

A calculation from paper to output

Paper step	Python instruction	Effect
Record pack count	packs = 4	Name holds 4
Multiply two quantities	total = packs * per_pack	Name holds product
Show the answer	print(total)	Displays the value

Assign per_pack before the multiplication. This diagram summarises the steps; the complete script below includes that assignment.

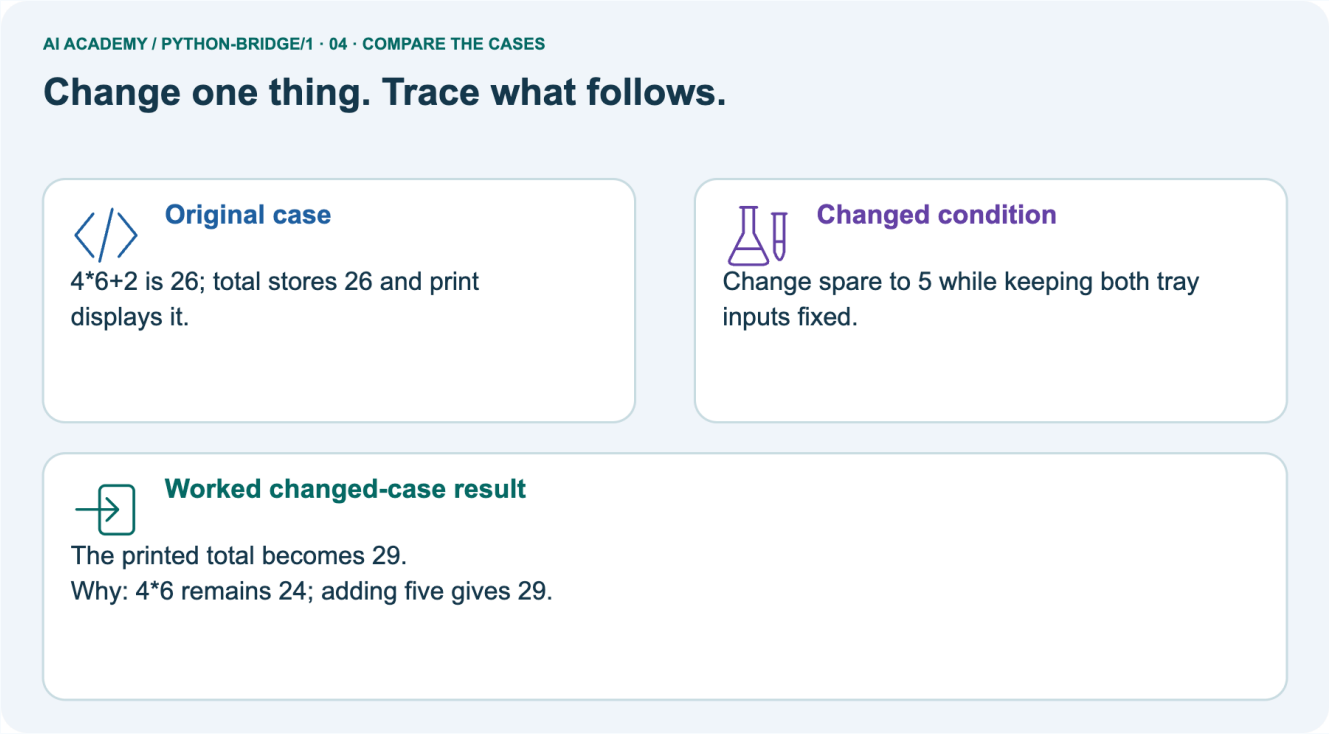
Follow the labelled picture with the matching explanation. Open the vector diagram

A calculation from paper to **output**. Record pack count | packs = 4 | Name holds 4. Multiply two quantities | total = packs * per_pack | Name holds product. Show the answer | print(total) | Displays the value

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

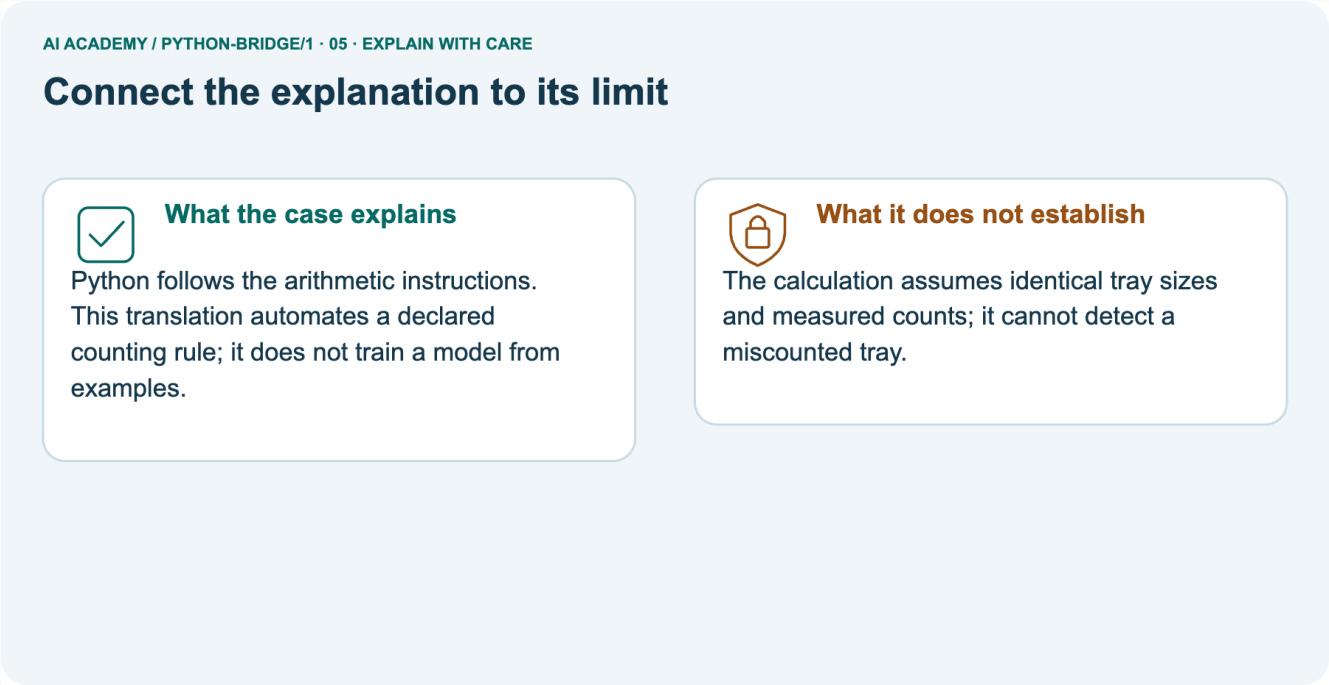
Original case: 4*6+2 is 26; total stores 26 and print displays it. Changed condition: Change spare to 5 while keeping both tray **input**s fixed. Expected result: The printed total becomes 29.

Reason: 4*6 remains 24; adding five gives 29.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

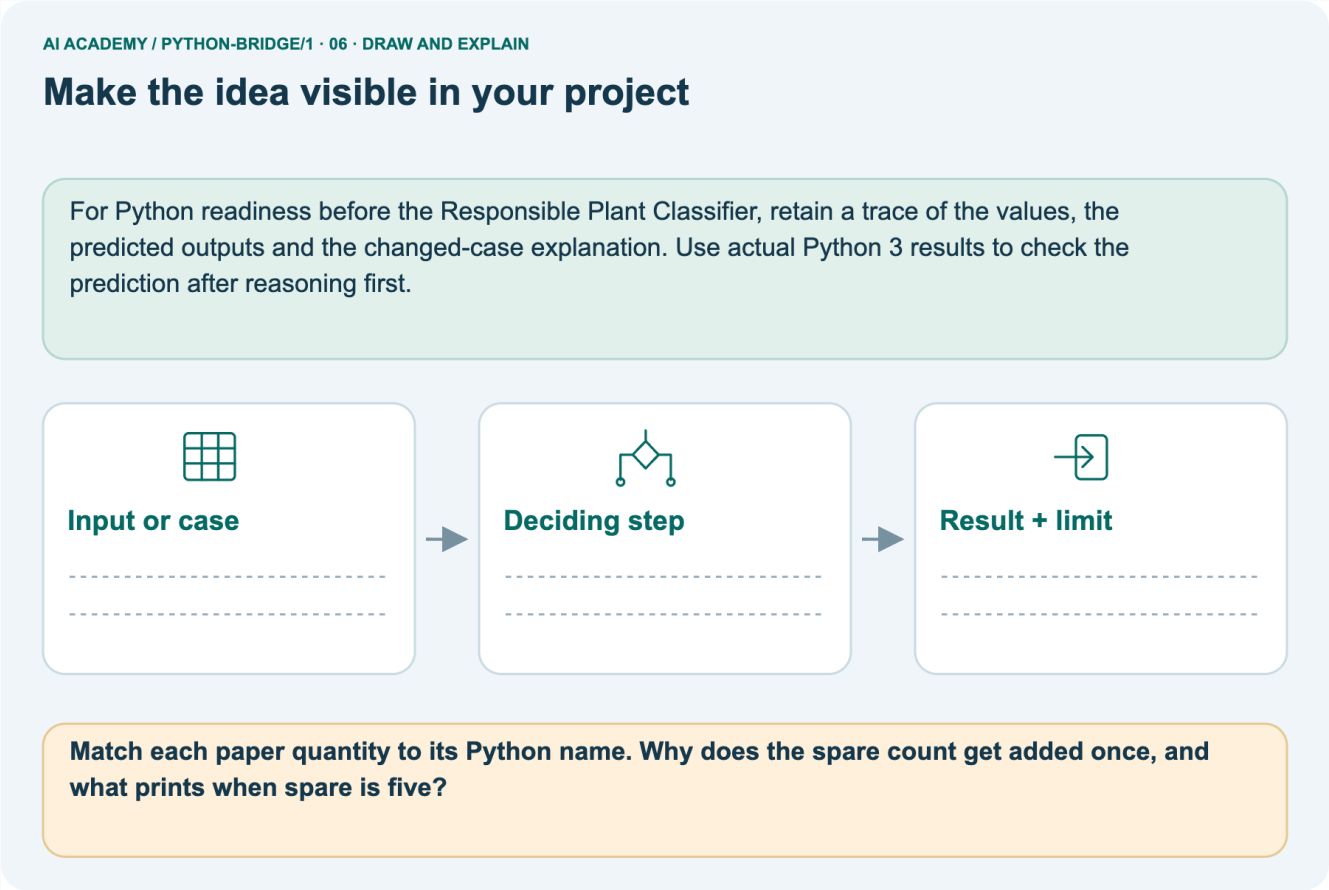
Python follows the arithmetic instructions. This translation automates a declared counting rule; it does not train a model from examples.

Limit: The calculation assumes identical tray sizes and measured counts; it cannot detect a miscounted tray.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Match each paper quantity to its Python name. Why does the spare count get added once, and what prints when spare is five?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.